

Grade 12 Physics

Mechanics: Kinematics · Dynamics · Momentum & Impulse

CAPS / IEB Curriculum • Comprehensive Study Notes

Contents

Topic	Sub-topics Covered
1. Kinematics	Equations of motion · Motion graphs · Projectile motion
2. Dynamics	Newton's three laws · Friction · Tension · Pulleys · Inclined planes
3. Momentum & Impulse	Momentum · Impulse · Conservation · Elastic & inelastic collisions

1 Kinematics

1.1 Equations of Motion (constant acceleration)

The following four equations apply whenever acceleration is constant. Identify the three known variables and choose the equation that contains your unknown.

Symbol	Quantity	SI Unit
u	Initial velocity	m/s
v	Final velocity	m/s
a	Acceleration	m/s ²
t	Time	s
s	Displacement	m

$$v = u + a t$$

$$s = u t + \frac{1}{2} a t^2$$

$$v^2 = u^2 + 2 a s$$

$$s = \frac{1}{2} (u + v) t$$

Tip: List the five variables (u , v , a , t , s). Identify the three knowns and the one unknown, then select the equation that contains exactly those four variables.

1.2 Motion Graphs

Graph type	Gradient (slope)	Area under graph
Displacement-time (s-t)	Velocity	—
Velocity-time (v-t)	Acceleration	Displacement
Acceleration-time (a-t)	—	Change in velocity (Δv)

On a **v-t graph** the area between the curve and the time-axis equals displacement. Areas *below* the time-axis represent motion in the negative direction.

1.3 Projectile Motion

A projectile moves under gravity alone (air resistance neglected). Treat the two directions **independently**:

Direction	Acceleration	Velocity
Horizontal (x)	0 m/s ² (no horizontal force)	Constant: $v(x) = u(x)$
Vertical (y)	$g = 9.8 \text{ m/s}^2$ downward	Changes: use equations of motion

- Time of flight – use vertical equations; horizontal time equals total vertical time.
- Maximum height – at the peak, $v(y) = 0$; solve for t then substitute.
- Range – horizontal distance = $u(x) \times \text{total time of flight}$.
- At launch angle θ from horizontal: $u(x) = u \cos\theta$, $u(y) = u \sin\theta$.

2 Dynamics (Newton's Laws)

2.1 First Law – Law of Inertia

Newton's First Law

An object remains at rest or continues moving at constant velocity in a straight line unless acted upon by a net (resultant) external force.

Inertia is the tendency of an object to resist changes to its state of motion. Greater mass means greater inertia.

2.2 Second Law – Law of Acceleration

Newton's Second Law

The net force acting on an object equals the product of the object's mass and its acceleration. The acceleration is in the same direction as the net force.

$$F(\text{net}) = m a$$

Draw a **free-body diagram** for every object. Sum all forces (taking a positive direction) to find $F(\text{net})$, then apply $F(\text{net}) = ma$.

2.3 Third Law – Action & Reaction

Newton's Third Law

For every action force there is an equal in magnitude but opposite in direction reaction force. The two forces act on *different* objects and cannot cancel each other.

2.4 Applications

Application	Key Relationship	Notes
Friction	$F(f) = \mu N$	$\mu(s)$ static $\geq \mu(k)$ kinetic; N = normal force
Tension (string)	T = force transmitted through rope	Massless string: same T throughout
Atwood machine (pulley)	$F(\text{net}) = (m_1 - m_2) g$	Apply $F = ma$ to each mass separately
Inclined plane	$F(\blacksquare) = mg \sin\theta$; $F(\perp) = mg \cos\theta$	Resolve weight along & perpendicular to slope

On an incline, friction acts up the slope when the object slides down, and down the slope when the object is pushed up. The normal force equals $mg \cos\theta$ (not mg) on a tilted surface.

3 Momentum and Impulse

3.1 Momentum

Momentum is a **vector** quantity with the same direction as the object's velocity.

$$p = m v \text{ (unit: kg}\cdot\text{m/s)}$$

A change in momentum requires a net force (Newton's Second Law in its general form: $F(\text{net}) = \Delta p / \Delta t$).

3.2 Impulse

$$\text{Impulse} = F(\text{avg}) \times \Delta t = \Delta p = m \Delta v$$

(unit: N·s or kg·m/s)

On a *force-time graph* the area under the curve equals the impulse (change in momentum). A longer collision time means a smaller average force for the same momentum change – the principle behind crumple zones and air bags.

3.3 Conservation of Momentum

Law of Conservation of Momentum

In an isolated system (zero net external force), the total momentum before an interaction equals the total momentum after the interaction.

$$\Sigma p(\text{before}) = \Sigma p(\text{after})$$

Momentum is a vector. Assign a positive direction and keep signs consistent throughout the calculation.

3.4 Types of Collision

Type	Momentum Conserved?	Kinetic Energy Conserved?	Example
Elastic	Yes	Yes	Ideal billiard balls, gas molecules
Inelastic	Yes	No – lost as heat / sound / deformation	Car crash, clay hitting wall
Perfectly inelastic	Yes	No – maximum KE lost	Objects that stick together after collision

Elastic collision kinetic-energy test:

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Calculate total KE before and after. If equal the collision is elastic; if the final KE is less it is inelastic.

Quick-Reference Formula Sheet

Formula	Name	Variables
$v = u + at$	1st equation of motion	v, u, a, t
$s = ut + \frac{1}{2} a t^2$	2nd equation of motion	s, u, a, t
$v^2 = u^2 + 2as$	3rd equation of motion	v, u, a, s
$s = \frac{1}{2} (u + v) t$	4th equation of motion	s, u, v, t

Formula	Name	Variables
$F(\text{net}) = ma$	Newton's Second Law	F, m, a
$F(f) = \mu N$	Friction force	F(f), μ , N
$p = mv$	Momentum	p, m, v
Impulse = $F(\text{avg}) \Delta t = \Delta p = m\Delta v$	Impulse-momentum theorem	F, Δt , Δp , m, Δv
$\Sigma p(\text{before}) = \Sigma p(\text{after})$	Conservation of momentum	p